Reflective Compass Control

Self-Aware IFS/ATP Architecture, Proof-Compass Synthesis, Federated Control, Formal Results, and a Preregistered Experimental Program

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Scientific status. This document proposes a research architecture and an experiment sequence. The formal results below are proved only under the explicit hypotheses stated. Empirical claims are restricted to previously frozen experiments and preregistrations. The verifier remains sovereign: reflective, learned, or federated control may change search order, but it may not manufacture object-level theoremhood.

Central idea. Treat the iterated-function-system (IFS) settlement process itself as an automated theorem prover (ATP), give that ATP a small certified self-theory, and use its self-knowledge to control a second-order proof compass. The resulting machine does not gain new axioms or a privileged truth oracle. It gains a formally auditable way to decide when its own first-order navigation should be trusted, when a bounded structural escape is justified, when verified information from peer provers should be imported, and when complete fallback should resume.

Research draft reconstructed as a LaTeX-native edition from the Version 0.1 report, with a new federation extension and corrected table layout.

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1. Executive Summary

The research program has two linked but separable objectives. Objective I is operational: make the IFS/ATP formally self-aware enough that derived facts about its own search can improve navigation. Objective II is foundational: determine whether the same construction can be certified as a non-degenerate formally self-aware ATP in the sense developed in *Depths of a Simulation and Formalized Self-Awareness*.

Proof-compass theory supplies the bridge. The first-order compass ranks legal proof successors. A new second-order or meta-compass ranks safe control policies using the ATP's certified self-state. A federation layer can additionally expose replay-verifiable lemmas, certificates, and calibration summaries from peer ATPs without allowing peer consensus to replace local verification.

Object layer	formal quotient → solved-DAG density → proof compass → candidate ordering
Reflective layer	certified self-state → meta-compass → safe controller / IFS-map selection
Federation layer	verified peer artifacts → replay checking → federated certified context
Safety layer	complete fallback → native verifier → accepted certificate

The strongest experimental question is therefore not simply whether "self-awareness helps." It is whether a compact, pre-outcome self-state contains information that predicts which safe search policy will win on this episode. That question admits a clean oracle-headroom analysis before a new meta-controller is trained. The analogous federation question is whether certified cross-node information supplies additional predictive or search value beyond what an isolated prover already possesses.

2. Formal Setting

Let D be the IFS/ATP. At stage m, write its observable state as the pair

$$S_m = (O_m, \Sigma_m),$$

where O_m is the object-level verified proof-search state and Σ_m is a set of certified, stage-indexed self-descriptive statements about D. Stage indexing is important because SIC states are cumulative: a statement such as "eight failures had occurred by stage m" remains true as a historical statement even after later progress.

$$\begin{aligned}
\text{State at stage } m &: S_m = (O_m, \Sigma_m),\\
\text{First-order compass} &: K_1(v, u) \text{ ranks legal proof successors } u,\\
\text{Second-order compass} &: K_2(\Sigma_m, a) \text{ ranks safe control actions } a,\\
\text{Selected action} &: a_m \in \arg\min_a \widehat{\text{cost}}(a \mid \Sigma_m),\\
\text{Evolution} &: S_{m+1} = T_{a_m}(S_m).
\end{aligned}$$

Examples of self-state formulas include:

- $\text{FailurePrefix}(D, m, 8)$: the first eight baseline candidates have failed by stage m.

- LocalDensity$(D, m, \widehat{p})$: the training-only solved-DAG density estimate near the current quotient state is $\widehat{p}$.
- EscapeAdvantage(D, m, c, δ): outsider c exceeds the strongest remaining local structural score by δ.
- CompassTension(D, m, c, d): structural evidence for c exceeds its normalized ordinary-compass score by d.
- FallbackComplete(D): the controller preserves the complete finite fallback schedule.

3. Density as Self-Knowledge About Compass Competence

Under the existing quotient-density one-hit model, let $p(q)$ be the probability mass of solved training states in a local quotient neighborhood $B(q)$, let n be the number of IID solved training states, and let K be the candidate count. The probability of at least one local training hit is

$$\rho(q) = 1 - (1 - p(q))^n.$$

Under local quotient constancy, one-hit transfer, and random ordering when $B(q)$ is unseen, the expected rank is

$$\mathbb{E}[R \mid q] = 1 + \frac{K-1}{2}(1 - p(q))^n,$$

and the corresponding expected gain over uninformed random navigation is

$$G(p) = \frac{K-1}{2}\left[1 - (1-p)^n\right].$$

Thus $\rho(q)$ can be interpreted as an analytically motivated measure of local support for the current compass. The reflective controller can combine this prior competence estimate with observed failures. A high-density region followed by eight failures is qualitatively different from eight failures in a region where the compass had little training support to begin with.

4. Formal Results Available Before New Experiments

Theorem 4.1 (Reflective-Control Conservativity)**.** *Suppose a reflective extension D^* of a sound rule-driven ATP D satisfies: (i) reflective rules alter only scheduling, ranking, beam width, quotient representative choice, or selection among already admissible search maps; (ii) every object-level theorem admitted to the verified bank passes the same sound inference or verifier condition as in D; (iii) reflection does not add object-level axioms; and (iv) complete fallback is fair and eventually covers every candidate covered by D. Then D^* proves no new object-level theorem merely by reflection, and every finite target eventually reachable by D remains eventually reachable by D^*. Reflection may change hitting time but not the verifier-defined object theorem set.*

Proof. Soundness follows from (i)–(iii): every admitted object theorem has exactly the same admissibility condition as before. Coverage follows from (iv): changing priority does not delete any candidate from the eventual schedule. Therefore reflection can alter transaction count or verifier calls without creating theoremhood. $\square$

Theorem 4.2 (No-Free-Reflection)**.** *Let X be the non-reflective observable feature state, let $Z = f(X)$ be a deterministic reflective encoding of those same features, and let the original controller class Π_X be closed under composition with f. Then allowing policies to read Z rather than X alone cannot improve the population optimum solely because the features were rewritten as self-descriptive formulas.*

Proof. Every Z-policy π_Z induces the X-policy $x \mapsto \pi_Z(f(x))$, which belongs to Π_X by closure. Hence the best achievable risk using Z is already achievable using X. Any empirical gain from a reflective controller must therefore come from new information, a different policy class, improved regularization or auditability, or a genuinely different control constraint–not from renaming old information as self-knowledge. □

Theorem 4.3 (Reflective-Control Decomposition)**.** *Let B be the expected cost of the best fixed safe base policy on a frozen episode distribution, let O be the expected cost of an oracle that selects the best safe base policy separately for each episode, and let R_Z be the cost of a learned reflective selector using only pre-outcome self-state Z. Then*

$$B - O = (B - R_Z) + (R_Z - O).$$

The term $B - O$ is total policy-switching headroom, $B - R_Z$ is captured headroom, and $R_Z - O$ is residual selection regret.

Proof. Add and subtract R_Z. □

Corollary 4.4 (Oracle-Headroom Gate)**.** *If $B - O$ is below a preregistered practical threshold, then no selector restricted to the same safe base-policy family can yield a practically large improvement over B. A small oracle gap is therefore a principled stop condition before fitting a meta-controller.*

Theorem 4.5 (Information Monotonicity)**.** *Let Y be episode outcome cost, X the ordinary observable state, and Z additional certified self-state. For any loss L, the Bayes risk using (X, Z) cannot exceed the Bayes risk using X alone, because a decision rule may ignore Z.*

Proof. The class of measurable decision rules based on X embeds in the class based on (X, Z) by ignoring Z. Taking the infimum over the larger class cannot increase risk. □

Theorem 4.6 (Finite Second-Order Compass Learnability)**.** *Fix a finite family of safe meta-policies $\mathcal{A}$, each mapping certified self-state into one of finitely many verifier-safe control actions. Suppose training episodes are IID, all per-episode policy losses lie in $[0, 1]$, and the policy family is fixed before confirmatory evaluation. Then empirical risk minimization over $\mathcal{A}$ obeys the finite-class bound*

$$\Pr\left(\sup_{a\in\mathcal{A}} |\widehat{R}(a) - R(a)| > \varepsilon\right) \leq 2|\mathcal{A}|e^{-2N\varepsilon^2}.$$

Hence for

$$N \geq \frac{\log(2|\mathcal{A}|/\delta)}{2\varepsilon^2},$$

all policy risks are simultaneously within ε of their empirical estimates with probability at least $1 - \delta$.

Proof. Apply Hoeffding's inequality to each fixed policy and a union bound over the finite class. □

5. Oracle-Headroom Analysis Before Training a Meta-Compass

The first computation after DATA 4.1-F completes should not be a new neural model. It should be an oracle table. For each episode i, record the cost $C_i(a)$ of every safe base policy a in a preregistered set $\mathcal{A}$. Compute

$$B = \min_{a\in\mathcal{A}} \frac{1}{N}\sum_i C_i(a), \qquad O = \frac{1}{N}\sum_i \min_{a\in\mathcal{A}} C_i(a).$$

If $B - O$ is tiny, there is little policy-switching value to learn. If it is substantial, then inspect whether certified self-state Z_i predicts which policy is near-optimal.

This separates three questions that are often conflated:

1. Do the base policies differ meaningfully by episode?
2. Does self-state predict those differences?
3. Can a frozen meta-policy capture enough of that signal without damaging fallback coverage?

A second-order compass should be trained only if the first two questions have affirmative evidence on development data.

6. Experimental Controls and Preregistration Discipline

- Freeze train/tune/confirm splits before computing confirmatory outcomes.
- Compute density from training solved-DAGs only; never from held-out proof edges or outcomes.
- Freeze candidate oceans across all arms within an experiment.
- Preserve paired evaluation: every controller sees the same target and same candidate ocean.
- Use a feature-matched external controller that performs the same arithmetic without self-descriptive syntax. Its rankings should match the reflective controller exactly when the feature set and policy are identical.
- Add a shuffled-self-state negative control. Shuffle density/surprise information across episodes while preserving marginal distributions. A genuine information-based reflective advantage should collapse or sharply weaken.
- Add a distribution-shift stress test because a learned compass can become an anti-compass under severe shift. Reflection should be evaluated as a fallible control signal, not as an oracle.
- Choose sample size from a preregistered minimal practical effect and prior paired-rank variance. Do not choose the confirmatory n after inspecting the reflective treatment's own outcomes.
- Report mean rank/verifier calls, MRR, median, p90, top-k rates, intervention frequency, improvement/harm counts, gain tails, and paired interval estimates. Do not collapse creativity to one scalar.
- For federation experiments, partition targets and training artifacts so a node cannot receive held-out outcome information through a peer. Every imported object claim must carry provenance and be locally replay-verifiable.

7. Decision Criteria for Advancing the Architecture

Question	**Advance when...**	**Stop / revise when...**
Is there policy-switching opportunity?	Oracle headroom $B-O$ exceeds a preregistered practical threshold.	$B-O$ is negligible; a meta-controller cannot buy much.
Does self-state contain useful information?	$B-R_Z$ is positive and stable; shuffled-self-state control loses the gain.	Performance survives shuffling or is no better than the best fixed policy.
Does reflection add value beyond syntax?	New certified information or safe policy selection improves outcomes.	Only a relabeling of old features is responsible; No-Free-Reflection applies.
Is it safe?	Same verifier, same theorem set, complete fallback, trace conformance.	Any action deletes fallback coverage or admits unverified object claims.
Is certification plausible?	Derived self-identification, provability soundness, non-degenerate object search, faithful SIC trace.	Self-theory is handed, dominates the whole formula space, or implementation/refinement cannot be certified.
Does federation add value?	Verified peer information improves hitting time or resource use beyond the best isolated node.	Gains disappear under provenance-preserving shuffle, or rely on unverified/contaminated information.

8. Proposed Architecture: DATA 4.2-RC

A provisional name for the integrated architecture is DATA 4.2-RC – Reflective Compass Control. It extends quotient-density compass navigation without replacing it:

formal quotient
↓
solved-DAG density
↓
first-order proof compass
↓
certified self-model of current search
↓
meta-compass over safe controllers / IFS maps
↓
conservative exhaustive fallback
↓
native verifier

The essential research claim is modest but substantive: a formally self-aware ATP may be able to improve expected certificate-hitting time by using correct, bounded statements about its own present search competence to choose among safe navigation policies. The claim does not require consciousness, phenomenology, or an internal truth oracle.

9. Federation of Reflective ATPs

Federation is a natural extension once a single reflective node is verifier-safe. Let $D_1, \ldots, D_M$ be ATP nodes, each with its own native verifier, local object state O_i, certified self-state Σ_i, and complete fallback schedule. A federation does *not* make theoremhood a voting problem. Instead, nodes exchange only artifacts that another node can independently check: proof certificates, verified lemmas, provenance records, quotient summaries, and preregistered calibration statistics.

For node i, let F_i denote the locally accepted federated context. A minimal safe update rule is

$$F_i^{(m+1)} = F_i^{(m)} \cup \{x : x \text{ arrived from a peer and passes node } i\text{'s replay-verification rule}\}.$$

A federated meta-compass may then condition on both local self-state and certified peer context:

$$K_F(\Sigma_i, F_i, a),$$

while object-level acceptance remains solely the responsibility of the node's native verifier.

Theorem 9.1 (Federated Conservativity Under Local Replay Verification)**.** *Suppose each node in a federation is sound; every imported object-level theorem or lemma is replay-verified locally under the receiving node's admissibility rules before entering its verified bank; federated information may alter only scheduling, ranking, admissible map selection, or resource allocation; no imported artifact adds an object-level axiom; and each node preserves fair complete fallback. Then federation may change proof-search hitting times and resource use, but it does not enlarge the verifier-defined object theorem set of any node merely by federation.*

Proof. An imported object claim is admitted only after satisfying the same local verifier condition that would be required for a locally generated claim. Federated control changes which admissible actions are attempted and when, but not the local theorem-admission predicate. Fair fallback preserves the original eventual coverage. Hence federation changes search dynamics, not verifier-defined theoremhood. □

A useful experimental decomposition is to compare four conditions on the same frozen target set: isolated nodes; nodes sharing only verified lemmas; nodes sharing verified lemmas plus calibration summaries; and a shuffled-peer-information negative control. This can distinguish ordinary parallelism from genuine federated compass value. A positive federation result should survive strict provenance checks and disappear or weaken when peer summaries are shuffled across episodes.

Potential federation benefits include diversity of search trajectories, cross-node reuse of certified lemmas, natural specialization of proof compasses, and more robust policy selection under distribution shift. The main risks are contamination, duplicated work, hidden dependence between nodes, and accidental promotion of peer heuristics into trusted truth. Those risks are controlled by local replay verification, explicit provenance, held-out partitions, and verifier sovereignty.

10. Work That Can Be Completed Before Compute Is Available

- Write the abstract two-sorted SIC/ATP definition and its trace semantics.

- Formalize the reflective self-state grammar and stage-indexed predicates.
- Prove the conservativity, no-free-reflection, control-decomposition, information-monotonicity, and second-order learnability results in a standalone technical note.
- Specify the exact conformance experiment so its expected result is equality with DATA 4.1-F.
- Specify the density estimator and pre-outcome surprise calibration without touching future confirmation labels.
- Define the finite meta-policy class and the oracle-headroom stopping rule.
- Write the full preregistration skeleton, leaving only quantities that must be frozen from the completed 4.1-F result or power calculation.
- Design the implementation trace format required for later SIC/refinement certification.
- Specify a federation protocol: certificate format, provenance schema, peer replay rules, contamination firewall, and the exact information allowed to cross node boundaries.

11. References / Internal Research Basis

1. Tenneson, B. *Depths of a Simulation and Formalized Self-Awareness*, Merged Edition, Version 1, July 27, 2026. Relevant sections: SIC/ATP definitions; self-certification and Picard Protocol; self-theories and levels; non-degeneracy; reflection-closure machine.
2. *Proof Compass Theory: Learnability, Shrinkage, and Control*, August 15, 2026. Relevant topics: compass learnability, controlled shrinkage, fallback, adaptive compass control, distribution-shift failure modes.
3. *Compass Theorems and Physics-Density Limits*, August 17, 2026. Relevant results: Compass Learnability, Navigation Advantage, Quotient-Density Calibration, Density-Conditional Navigation Advantage.
4. *DATA 4.2: Quotient-Density Compass and Conservative Fallback*, August 17, 2026. Relevant architecture: formal quotient, quotient-density training, compass-ordered search, conservative fallback, verifier sovereignty.
5. *DATA 4.1-F Evidence-Selective Structural Escape – Preregistration*, August 19, 2026. Frozen baseline for structural escape, thresholds, ablations, confirmation endpoints, and complete finite fallback.

Bottom line. The best next step is not to make the ATP maximally reflective. It is to make it usefully reflective: derive a small set of correct statements about its own current navigation competence, measure whether those statements contain policy-switching information, and exploit that information only through verifier-safe control with complete fallback. Federation extends the same discipline across multiple ATPs: peers may share certified evidence and calibration, but local verification remains sovereign.